CIS 678 Machine Learning

Special Topics (part 2)

Outline

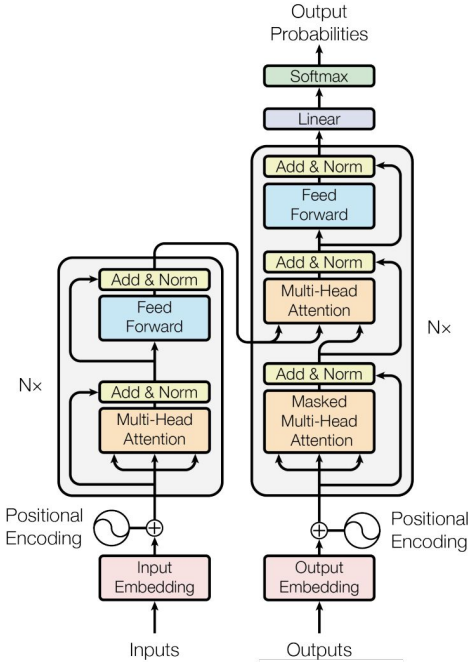
Introduction to

- Recurrent Neural Networks (RNN)
- **Transformers**
- Generative Modeling
- Foundational Models

Transformers

Examples:

- *Encoder decoder pair*
- *GPT*
- *BERT*



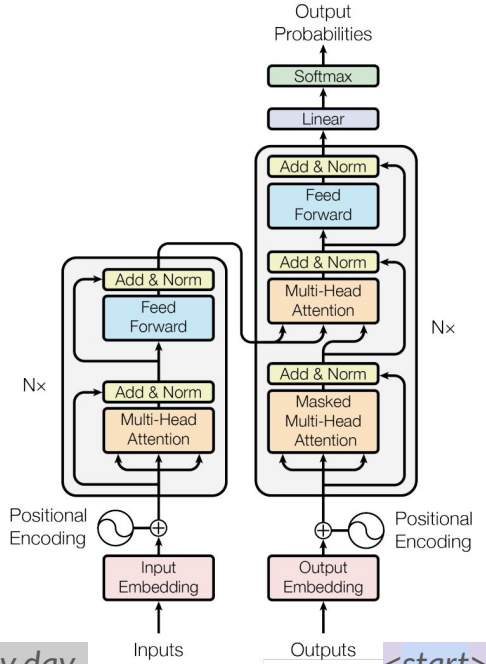
Example Task: ***Machine Translation***

Transformers

Examples:

- *Encoder decoder pair*
- *GPT*
- *BERT*

Une journee nuageuse



A cloudy day

<start> une journee nuageuse

Example Task: ***Machine Translation***

Transformers

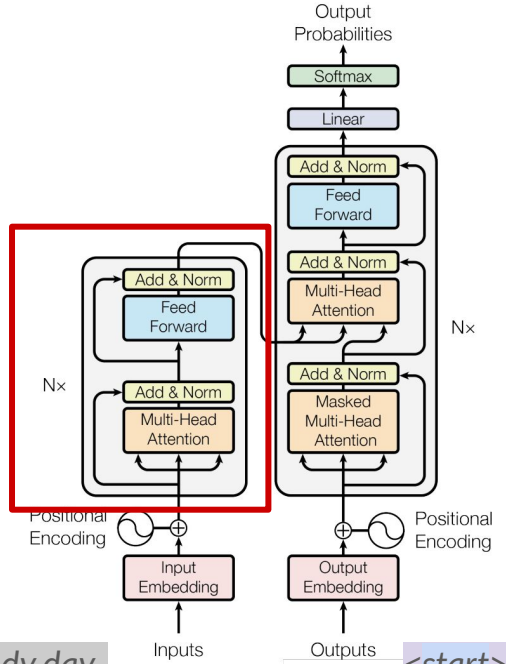
Examples:

- *Encoder decoder pair*
- *BERT Encoder*
- *GPT Decoder*

BERT

Encoder

Une journee nuageuse



A cloudy day

<start> une journee nuageuse

Example Task: ***Machine Translation***

Transformers

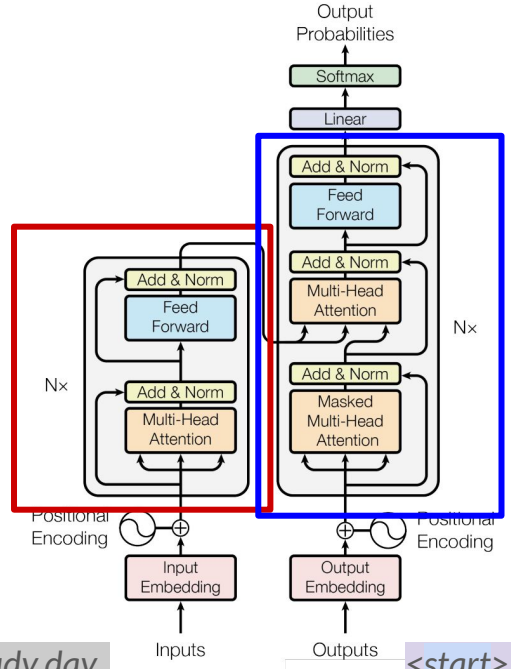
Examples:

- *Encoder decoder pair*
- *BERT Encoder*
- *GPT Decoder*

BERT

Encoder

Une journee nuageuse



GPT

Decoder

Example Task: ***Machine Translation***

A cloudy day

<start> une journee nuageuse

Transformers

Examples:

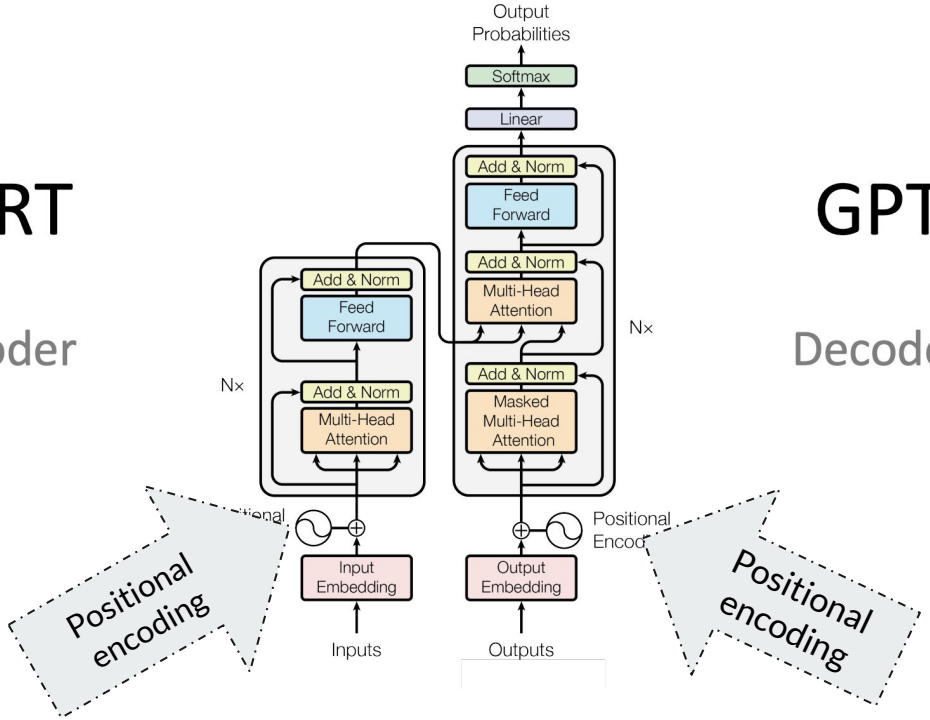
- *Encoder decoder pair*
- *BERT Encoder*
- *GPT Decoder*

BERT

Encoder

GPT

Decoder



A cloudy day

Day by day, the weather got cloudier.

Transformers

Examples:

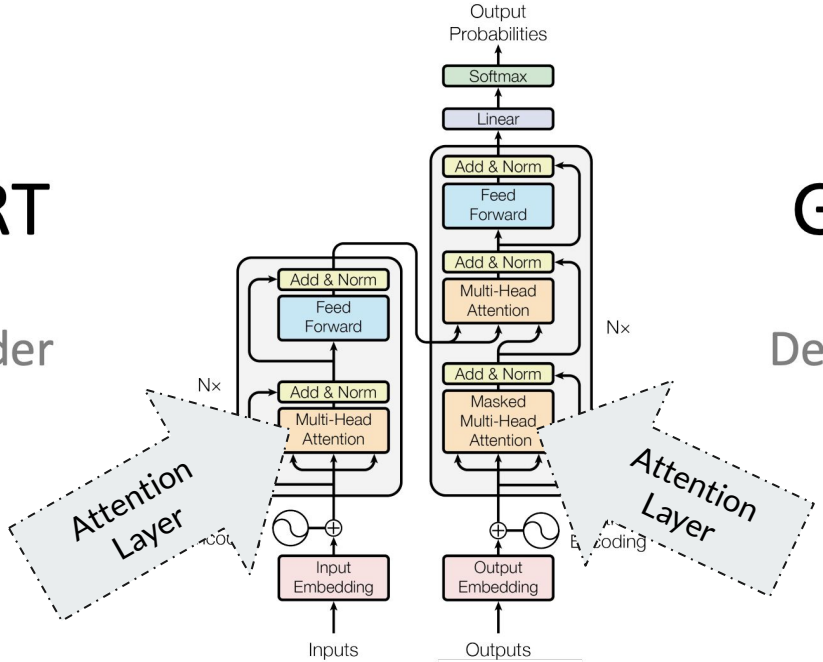
- *Encoder decoder pair*
- *BERT Encoder*
- *GPT Decoder*

BERT

Encoder

GPT

Decoder



The animal didn't cross the street because it was too tired

Transformers

Examples:

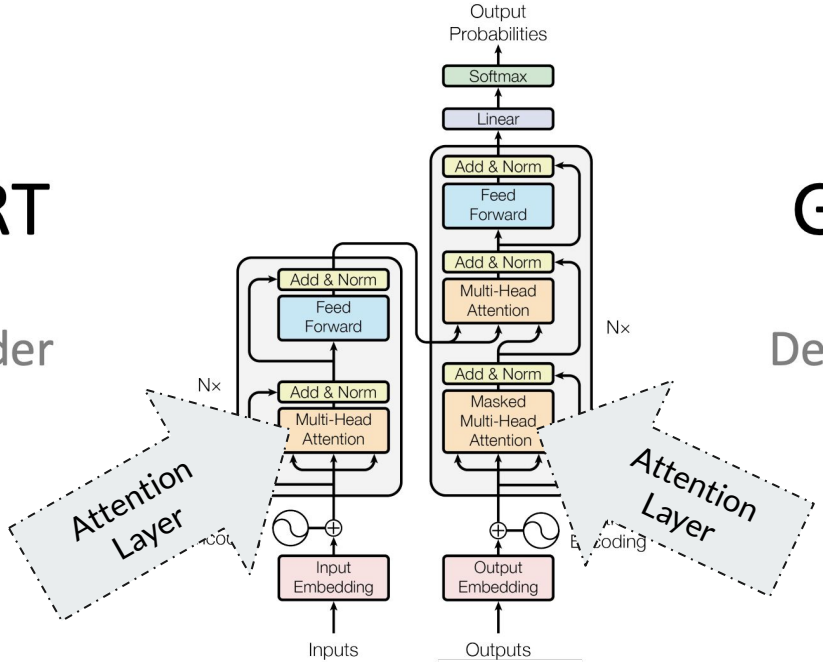
- *Encoder decoder pair*
- *BERT Encoder*
- *GPT Decoder*

BERT

Encoder

GPT

Decoder



The animal didn't cross the street because it was too tired

Transformers

Examples:

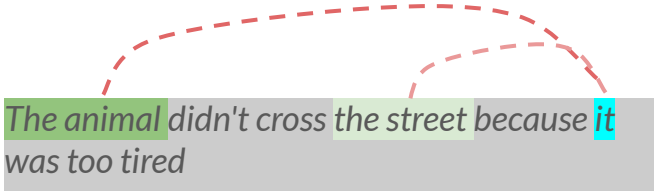
- *Encoder decoder pair*
- *BERT Encoder*
- *GPT Decoder*

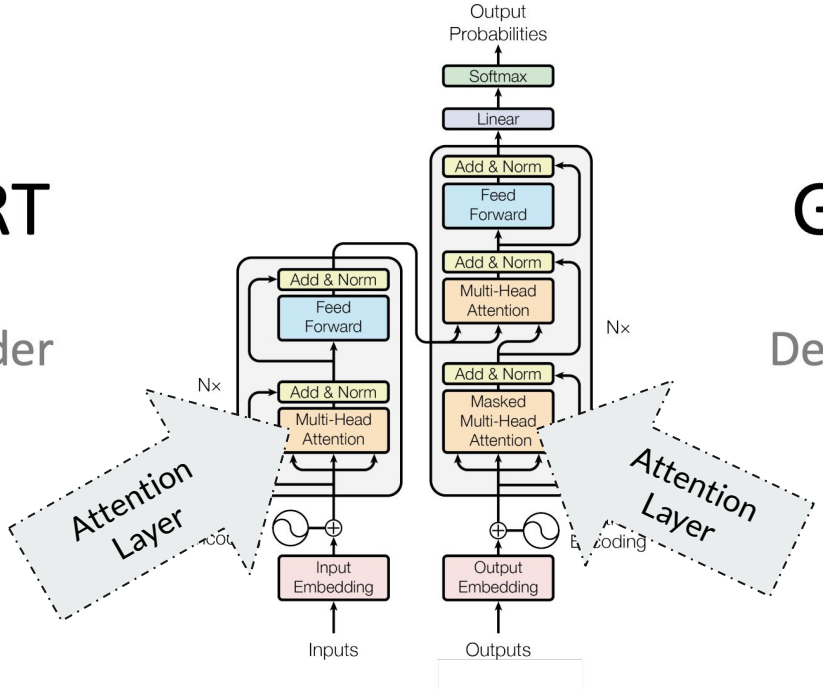
BERT

Encoder

GPT

Decoder





Transformers

Examples:

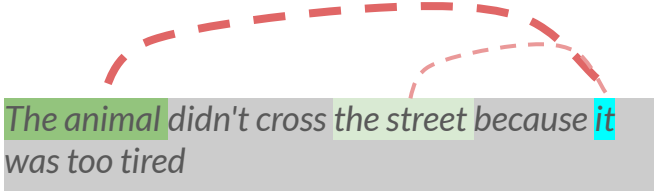
- *Encoder decoder pair*
- *BERT Encoder*
- *GPT Decoder*

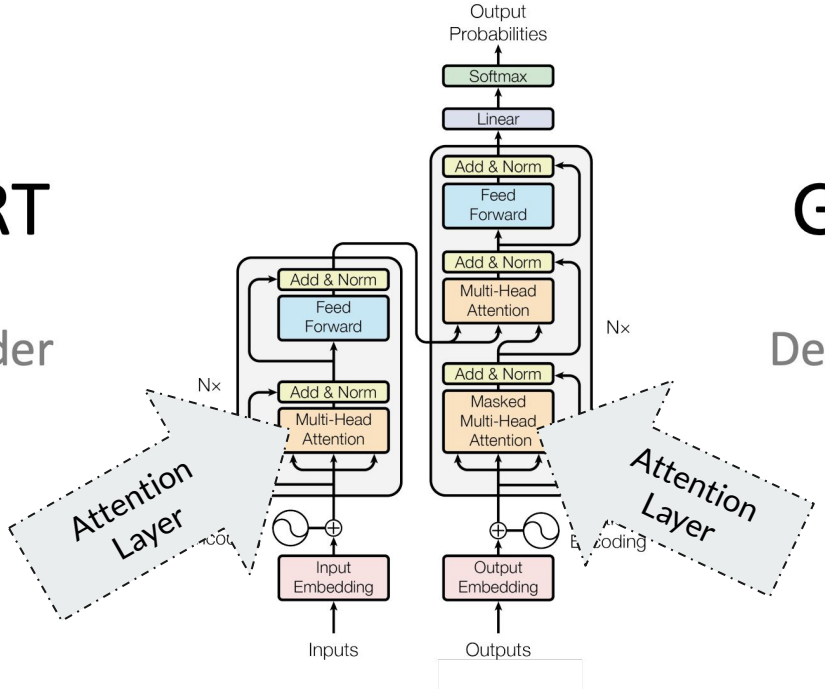
BERT

Encoder

GPT

Decoder





Transformers

Examples:

- *Encoder decoder pair*
- *BERT Encoder*
- *GPT Decoder*

BERT

Encoder

Une journee nuageuse

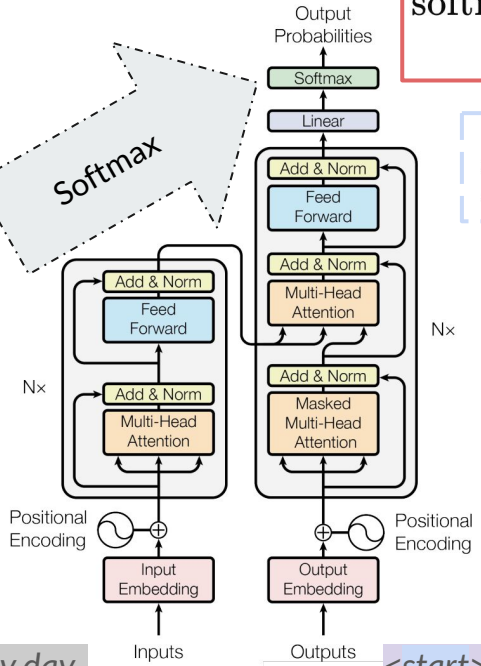
$$\mathrm{softmax}(z_i) = \frac{e^{z_i}}{\sum_{j=1}^{n} e^{z_j}}$$

Softmax

$$\frac{1}{1+e^{-z}}$$

Logistic

Decoder



A cloudy day

<start> une journee nuageuse

Example Task: ***Machine Translation***

QA